FoodLogAthl-218: Constructing a Real-World Food Image Dataset Using Dietary Management Applications

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Abstract

Food image recognition in real-world dietary management settings remains challenging due to the gap between curated web-crawled datasets and the noisy, casual photographs that users actually capture. We introduce FoodLogAthl-218, a benchmark dataset constructed from 6,925 real meal images containing 14,349 dish bounding boxes across 218 consolidated food classes, sourced from the dietary management application FoodLog Athl. Unlike existing food image datasets predominantly collected from the web, our dataset captures authentic dietary logging behavior from 513 anonymized users over an 18-month period (May 2023–October 2024), exhibiting natural class imbalance, diverse imaging conditions, and multi-dish compositions. We describe a six-step construction pipeline combining automated filtering with CLIP-based outlier detection and manual review, followed by label consolidation using GPT-4o. We define three benchmark tasks: closed-set dish classification, continual user-aware adaptation, and menu/context-aware per-dish classification. Extensive experiments with ten vision–language models reveal that the dataset poses significant challenges—the best model (GPT-4o) achieves only 42.4% Recall@1—and that user-specific fine-tuning and contextual descriptions offer promising directions for improvement. FoodLogAthl-218 provides a realistic testbed for advancing food image recognition toward practical dietary management applications.

1 Introduction

Accurate food image recognition is a key enabler for automated dietary management systems that help users track nutritional intake [3, 28]. While significant progress has been made on curated datasets such as Food-101 [6] and UEC-Food 256 [18], these benchmarks are predominantly constructed from web-crawled images that do not reflect the conditions encountered in real dietary logging applications. In practice, users capture meals under varying lighting, angles, and compositions, often with multiple dishes per image and visually similar food items from the same cuisine.

Several large-scale food datasets have been proposed in recent years [24, 27, 31], yet nearly all rely on web image search engines as their primary data source (Table 2). This introduces a systematic bias: web images tend to be well-composed, professionally photographed, and depict canonical presentations of dishes. In contrast, images from dietary management applications are casual photographs taken by everyday users primarily for record-keeping purposes, not for visual appeal. This domain gap limits the transferability of models trained on web-crawled data to real-world dietary applications.

To bridge this gap, we introduce **FoodLogAthl-218**, a food image dataset constructed entirely from logs of FoodLog Athl [28], a dietary management application used by athletes and health-conscious individuals. The dataset comprises 6,925 images containing 14,349 dish bounding boxes across 218 consolidated food classes, collected from 513 anonymized users between May 2023 and October 2024. Key characteristics that distinguish our dataset include: (1) authentic casual food photographs from real dietary logging, (2) natural long-tail class distribution reflecting actual eating patterns, (3) multi-dish images requiring per-dish recognition, and (4) temporal and user-level metadata enabling continual and personalized evaluation.

We propose three benchmark tasks that reflect practical challenges in dietary management: *closed-set dish classification*, which evaluates standard recognition performance; *continual user-aware adaptation*, which tests the ability to improve over time with user-specific data; and *menu/context-aware per-dish classification*, which investigates whether textual descriptions of dishes can guide recognition. We evaluate ten vision–language models spanning CLIP-based approaches [29, 32], large multimodal models (LMMs) [1, 5, 21, 22], and LoRA-based fine-tuning [15], providing comprehensive baselines for future research.

Our contributions are as follows:

- We construct FoodLogAthl-218, the first food image benchmark sourced entirely from a real dietary management application, with a rigorous six-step pipeline combining automated and manual quality control.
- We define three benchmark tasks that address practical needs in dietary management: closed-set classification, continual adaptation, and description-guided recognition.
- We provide extensive baseline evaluations showing that the dataset poses significant challenges for current models, with the best model achieving only 42.4% Recall@1, and demonstrate that user-specific adaptation and contextual information can meaningfully improve performance.

2 Related Works

2.1 Food Image Datasets

Food image recognition has been studied extensively over the past two decades. Early datasets such as PFID [8] focused on specific cuisines with limited scale. Joutou and Yanai [17] and Matsuda et al. [25] introduced multi-class food recognition benchmarks from Japanese cuisine. Food-101 [6] established a widely-used benchmark with 101 classes and 101,000 images crawled from foodspotting.com. UEC-Food 256 [18] extended coverage to 256 classes with bounding box annotations. More recent efforts have scaled to hundreds or thousands of classes: FoodAI-756 [31] covers 756 classes with 400,000 images, and Food2K [27] provides over one million images across 2,000 classes.

Despite this progress, virtually all existing datasets are collected from web sources (Table 2). UNICT-FD1200 [11] is a notable exception, using images from experiment participants, but remains small in scale. VegFru [14] and Food-500 Cap [24] further expanded coverage to vegetables, fruits, and captioned images, respectively. Our dataset is unique in sourcing data from a real dietary management

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application, capturing authentic user behavior rather than curated web content.

2.2 Large Multimodal Models for Food Recognition

Large multimodal models (LMMs) have demonstrated strong zero-shot capabilities across diverse visual recognition tasks [1, 22]. GPT-4o [1] and the Qwen-VL series [5] represent the current frontier of commercial and open-source LMMs, respectively. The LLaVA family [21, 22] provides efficient open-source alternatives. For food recognition specifically, Imajuku et al. [16] explored LMM-based approaches on FoodLog data. CLIP [29] and its successors such as EVA-CLIP [32] enable zero-shot classification through vision–language alignment. These models can be applied to food recognition either through direct zero-shot classification or through a description-then-match pipeline, as we explore in this work.

2.3 Continual and Personalized Food Recognition

Real-world dietary management requires models that adapt to individual users over time. Horiguchi et al. [13] demonstrated the value of personalized models for food recognition, showing that user-specific adaptation can significantly improve accuracy. Kim et al. [19] further explored boosting food recognition with continual learning from user-specific data. Parameter-efficient fine-tuning methods such as LoRA [15] make user-specific adaptation practical by reducing the computational cost of maintaining per-user models. The FoodLog system [3, 28, 35] has served as a platform for studying these personalized approaches. Our benchmark formalizes the continual adaptation task with temporal evaluation protocols that reflect realistic deployment scenarios.

3 FoodLogAthl-218 Dataset

3.1 Data Source

FoodLogAthl-218 is constructed from dietary logs collected through FoodLog Athl [28], a mobile application for dietary management used primarily by athletes and health-conscious individuals. Users photograph their meals and log them with timestamps. The application provides bounding box annotations for individual dishes within each image through its food detection system [2, 4]. Our dataset spans logs from 513 anonymized users collected between May 2023 and October 2024, preserving date and time metadata while removing all personally identifiable information. Figure 1 shows representative examples from the dataset, illustrating the diversity of imaging conditions, dish types, and compositions.

3.2 Construction Pipeline

We employ a six-step pipeline to transform raw dietary logs into a clean, well-structured benchmark dataset. Table 1 summarizes the statistics at each stage.

Step 0: Raw data. The initial dataset comprises 11,460 images with 61,414 bounding boxes spanning 5,889 class labels assigned by users and the application's recognition system.

Step 1: Remove samples without images. We remove bounding box entries where the corresponding image file is missing or corrupted, reducing the bbox count from 61,414 to 26,646 and classes from 5,889 to 3,167.

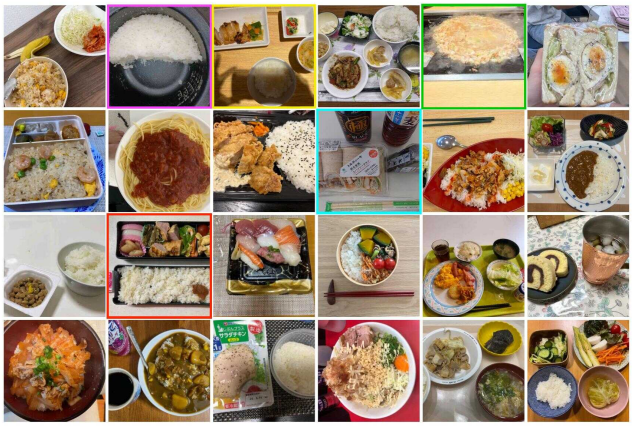

Figure 1: Example images from FoodLogAthl-218. The dataset contains casual meal photographs captured by real users of the FoodLog Athl dietary management application, exhibiting diverse lighting, angles, and multi-dish compositions.

Step 2: Remove samples without bounding boxes. Images lacking valid bounding box annotations are excluded, yielding 11,285 images with 26,065 bounding boxes across 3,071 classes.

Step 3: Remove rare classes. Classes with fewer than 10 samples are removed to ensure sufficient representation for evaluation, reducing to 9,121 images, 21,944 bounding boxes, and 1,180 classes.

Step 4: CLIP-based outlier filtering. We extract features using CLIP ViT-B/16 [29] for all cropped dish images and compute the centroid for each class. Samples whose feature distance exceeds 1.5 standard deviations from their class centroid are flagged as potential outliers and removed, reducing to 7,537 images and 17,513 bounding boxes while preserving all 1,180 classes.

Step 5: Manual review. Trained annotators review flagged and borderline cases. To maintain image-level consistency, we apply an all-or-none policy: if any bounding box in an image is deemed invalid, the entire image is removed. This yields 6,925 images with 14,349 bounding boxes.

Step 6: Label consolidation. The original 1,180 classes contain many near-synonymous or overlapping labels (e.g., different romanizations of the same dish). We use GPT-4o [1] to propose class merges based on semantic similarity of label names, followed by manual refinement to produce the final 218 consolidated classes.

3.3 Class Distribution

The resulting dataset exhibits a pronounced long-tail distribution (Figure 2), consistent with natural eating patterns where staple items (e.g., rice, miso soup) appear frequently while specialized dishes are rare. The most frequent class contains over 800 samples, while many classes have fewer than 30. This imbalance reflects realistic conditions in dietary management and poses challenges for classification models that must handle both common and rare dishes.

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Table 1: Data Filtering Process and the Number of Images, Bounding Boxes (bboxes) and Classes.

Step	Operation	#images	(diff)	#bboxes	(diff)	#classes	(diff)
0	Raw data	11,460		61,414		5,889	
1	Remove samples w/o image	11,460	(0)	26,646	(-34,768)	3,167	(-2,722)
2	Remove samples w/o bounding box	11,285	(-175)	26,065	(-581)	3,071	(-96)
3	Remove classes w/ <10 samples	9,121	(-2,164)	21,944	(-4,121)	1,180	(-1,891)
4	Remove outliers based on image features	7,537	(-1,584)	17,513	(-4,431)	1,180	(0)
5	Manual review and remove invalid images	6,925	(-612)	14,349	(-3,164)	1,180	(0)
6	Merge similar classes and clean labels	6,925	(0)	14,349	(0)	218	(-982)

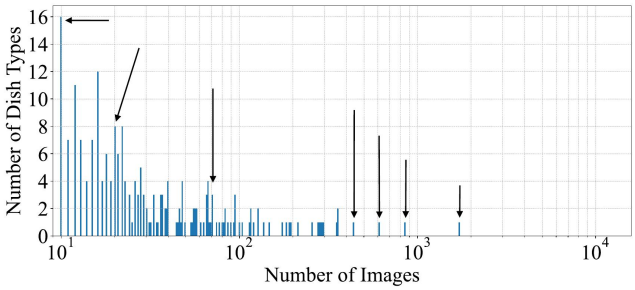

Figure 2: Class frequency distribution of FoodLogAthl-218. The dataset exhibits a natural long-tail distribution reflecting real-world dietary patterns, where common staple foods dominate while many specialized dishes appear infrequently.

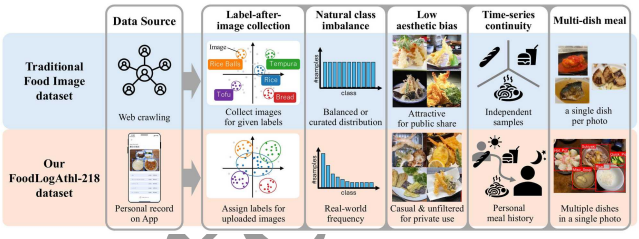

Figure 3: Feature space visualization comparing FoodLogAthl-218 with traditional web-sourced datasets. Our dataset exhibits greater intra-class diversity and inter-class overlap, reflecting the challenging nature of real dietary logging images.

3.4 Dataset Statistics and Comparison

Table 2 compares FoodLogAthl-218 with existing food image datasets. While our dataset is moderate in scale (218 classes, 14,349 bounding boxes), it is the only benchmark sourced from a real dietary management application. This unique provenance results in fundamentally different image characteristics: casual angles, inconsistent lighting, partial occlusions, and multi-dish compositions that are rare in web-crawled datasets.

To quantify these differences, we compute distribution metrics using features from CLIP ViT-L/14 [29], EVA-CLIP [32], and Swin Transformer [23] (Table 3). Compared to UEC-Food 256, FoodLogAthl-218 exhibits higher Trace values (CLIP: 81.70 vs. 71.21), indicating greater overall feature spread; higher Davies–Bouldin indices [9] (CLIP: 4.52 vs. 3.64), indicating less separable clusters; lower Silhouette scores [30] (CLIP: −0.021 vs. 0.034), indicating weaker cluster compactness; and lower Calinski–Harabasz scores [7] (CLIP: 27.51 vs. 74.05). These metrics consistently confirm that FoodLogAthl-218 presents greater intra-class diversity and inter-class overlap, making it a more challenging benchmark than traditional web-sourced datasets. Figure 3 visualizes the feature space distributions of both datasets.

4 Benchmark Tasks and Experimental Setup

We define three benchmark tasks that address distinct challenges in real-world dietary management, illustrated in Figure 4.

4.1 Task 1: Closed-Set Dish Classification

Given a cropped dish image, the model must predict the correct class label from the 218 predefined categories. This task evaluates the baseline recognition capability of current vision–language models in a zero-shot setting.

Evaluation protocol. For CLIP-based models (CLIP ViT-L/14, EVA-CLIP ViT-g), we compute cosine similarity between the image embedding and text embeddings of all 218 class names, ranking candidates by similarity. For LMMs (GPT-4o, Qwen2.5-VL-7B, Qwen2-VL-7B, Qwen2-VL-3B, LLaVA-OneVision-7B, LLaVA-v1.6-34B, LLaVA-v1.5-7B, LLaVA-v1.6-Mistral-7B), evaluated via lmms-eval [20], we prompt the model to generate a free-form textual description of the dish, then embed this description using intfloat/e5-large-v2 [33] and compute cosine similarity against embeddings of all class labels. This two-stage approach enables ranking-based evaluation (Recall@K) even for generative models.

Metrics. We report macro-averaged F1 score, and Recall@1, Recall@3, and Recall@5.

4.2 Task 2: Continual User-Aware Adaptation

This task evaluates whether models can improve over time by adapting to individual users' dietary patterns. We simulate a realistic deployment scenario where a model is periodically fine-tuned on accumulated user data.

Setup. We use Qwen2.5-VL-3B as the base model and apply LoRA fine-tuning [15] at weekly intervals. At each time step, the model is fine-tuned on all data available up to that point and evaluated on data from the subsequent period. We evaluate both a global model (trained on all users' data) and user-specific models (trained on individual users' data). Performance is reported as a weighted one-month moving average (±2 weeks) of classification accuracy.

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Table 2: Comparison of Datasets for Food Image Classification Tasks.

Year	Dataset Name	#classes	#samples	Cuisine Category	Data Source
2009	PFID [8]	101	4,545	Western	Fast Food Chains
2014	UEC-Food 256 [18]	256	25,088	International	**Web** + Original Database
2014	Food-101 [6]	101	101,000	Western	**Web** (foodspotting.com)
2015	UPMC Food-101 [34]	101	90,840	Western	**Web** (Google [12])
2016	UNICT-FD1200 [11]	1,200	4,754	International	Experiment Participants
2017	Vegfru [14]	25/292*	160,000	Vegetables, Fruits	**Web** (Google, ImageNet [10], etc.)
2019	FoodAI-756 [31]	152/756*	400,000	International	**Web** (Google, Bing, etc.)
2021	Food2K [27]	2,000	1,036,564	International	**Web** (Meituan [26])
2023	Food-500 Cap [24]	494	24,700	International	**Web** (Google, Bing)
2025	FoodLogAthl-218	218	14,349	International	**Dietary Management App**

* Hierarchically structured dataset, where n/m indicates m classes divided into n higher-level classes.

Table 3: Comparison of distribution metrics between FoodLogAthl-218 and UEC-Food 256 using feature extractors. Higher *Trace* and *Davies–Bouldin* indicate greater spread, while lower *Silhouette* and *Calinski–Harabasz* indicate less compact clusters.

	Trace (↑)			Silhouette (↓)			Davies–Bouldin (↑)			Calinski–Harabasz (↓)		
Dataset	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**	**CLIP**	**EVA-CLIP**	**Swin**
UEC-Food 256	71.21	45.09	264.00	0.0335	0.0577	-0.0064	3.64	3.02	4.11	74.05	115.80	49.17
FoodLogAthl-218 (Ours)	**81.70**	**49.92**	**282.57**	**-0.0210**	**0.0036**	**-0.0288**	**4.52**	**3.68**	**4.51**	**27.51**	**51.51**	**21.29**

4.3 Task 3: Menu/Context-Aware Per-Dish Classification

This task investigates whether providing textual descriptions of dish characteristics can improve recognition accuracy, simulating a scenario where menu information or contextual knowledge is available.

Setup. We use Qwen2.5-VL-7B and provide each dish class with a textual description of its visual characteristics, ingredients, and typical presentation. The model receives both the cropped dish image and the description as input, and must predict the correct class.

Metrics. We report F1 score, Precision@1 (P@1), and Recall@K (for $K \in \{1, 3, 5\}$).

Table 4: Classification performance on FoodLogAthl-218.

Model	**F1**	**R@1**	**R@3**	**R@5**
GPT-4o	**0.3732**	**0.4243**	**0.4991**	0.5221
Qwen2.5-VL-7B-Instruct	0.2349	0.3684	0.4609	0.4859
Qwen2-VL-7B-Instruct	0.1594	0.3656	0.4590	0.4781
Qwen2-VL-3B-Instruct	0.1386	0.2663	0.3250	0.3517
llava-onevision-qwen2-7b-ov	0.1546	0.3629	0.4497	0.4639
EVA-CLIP (ViT-g)	0.2465	0.2654	0.4897	0.5944
CLIP (ViT-L/14)	0.2393	0.2567	0.4885	**0.6000**
llava-v1.6-34b	0.1354	0.2432	0.3779	0.4063
llava-v1.5-7b	0.0991	0.2140	0.2721	0.2973
llava-v1.6-mistral-7b	0.0944	0.1932	0.2595	0.2831

5 Results

5.1 Closed-Set Classification (Task 1)

Table 4 presents the classification performance of ten models on FoodLogAthl-218. GPT-4o achieves the best F1 score (0.373) and Recall@1 (0.424), substantially outperforming all other models. Among open-source LMMs, Qwen2.5-VL-7B obtains the highest F1 (0.235) and competitive Recall@1 (0.368). Notably, the CLIP-based models (CLIP ViT-L/14 and EVA-CLIP) achieve the highest Recall@5 (0.600 and 0.594, respectively), indicating that the correct class frequently appears among the top-5 candidates even when top-1 precision is low.

The overall low performance—even the best model achieves only 42.4% Recall@1—confirms that FoodLogAthl-218 poses significant challenges. The LLaVA family models [21, 22] show relatively lower performance, with LLaVA-v1.6-Mistral-7B achieving only 19.3% Recall@1. This suggests that the description-then-match pipeline is sensitive to the quality and specificity of generated descriptions, and that larger or more capable LMMs produce more discriminative textual descriptions.

Figure 5 shows representative error cases, illustrating common failure modes including visually similar dishes from the same cuisine, unusual plating or lighting, and partially occluded dishes.

5.2 Distribution Analysis

The distribution metrics in Table 3 provide quantitative evidence for the difficulty of FoodLogAthl-218. Across all three feature extractors (CLIP, EVA-CLIP, Swin Transformer), our dataset consistently shows higher Trace and Davies–Bouldin values alongside lower Silhouette and Calinski–Harabasz scores compared to UEC-Food 256. This indicates that FoodLogAthl-218 has greater intra-class variance (dishes of the same type look more diverse) and weaker inter-class separation (different dishes look more similar). These properties directly stem from the real-world data source: users photograph the same dish type under widely varying conditions,

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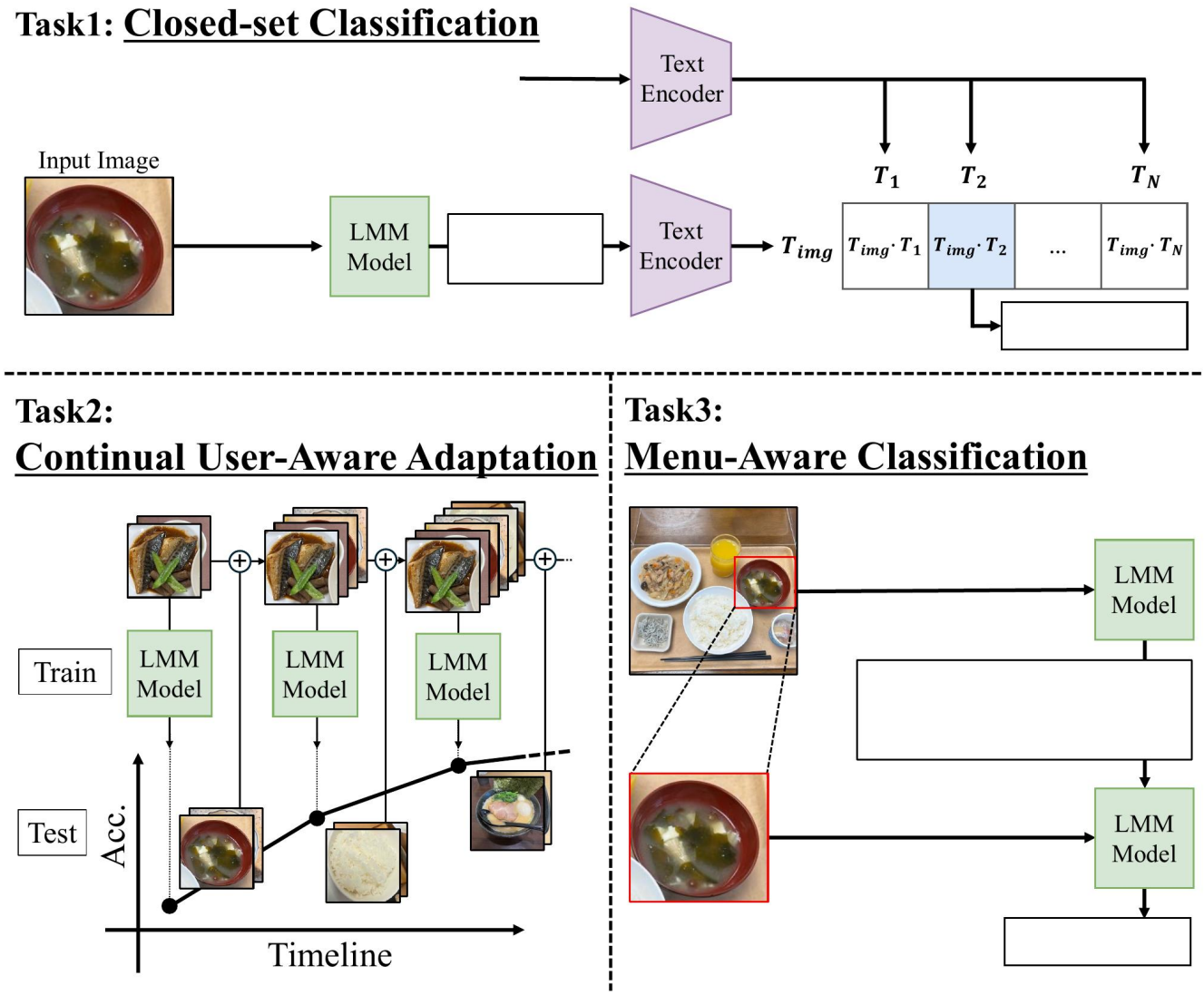

Figure 4: Overview of the three benchmark tasks defined on FoodLogAthl-218. Task 1: Closed-set dish classification evaluates zero-shot recognition. Task 2: Continual user-aware adaptation measures improvement through LoRA fine-tuning over time. Task 3: Menu/context-aware classification investigates the benefit of textual dish descriptions.

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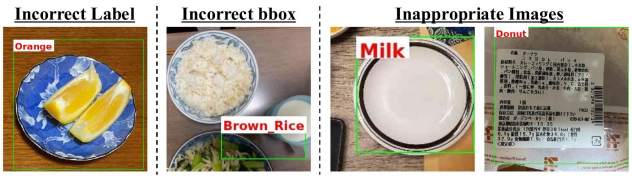

Figure 5: Representative error cases in closed-set classification. Common failure modes include visually similar dishes, unusual presentations, and partially occluded items.

and many Japanese and international dishes share similar visual elements (bowls of rice, noodle-based dishes, various stewed items).

5.3 Continual User-Aware Adaptation (Task 2)

Figure 6 shows the classification accuracy over time for the continual adaptation experiment across all users. The base Qwen2.5-VL-3B model without fine-tuning achieves approximately 20% accuracy. With LoRA fine-tuning on accumulated data, accuracy improves substantially, exceeding 50% as more training data becomes available. This demonstrates that continual adaptation is effective for real-world dietary data, where the model benefits from observing the specific dishes and presentation styles of the user population.

Figure 7 presents results for user-specific LoRA adaptation, focusing on a heavy-logging user (User A). The user-specific model achieves accuracy gains of approximately 50 percentage points over the base model, dramatically outperforming the global model. This result highlights the potential of personalized adaptation in dietary

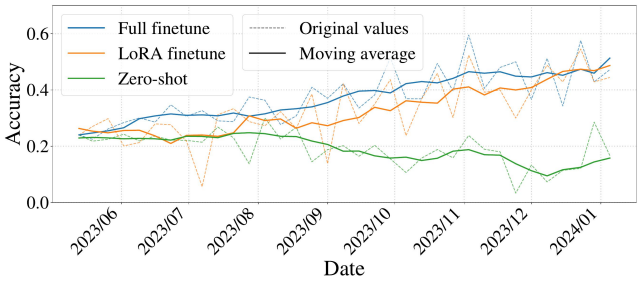

Figure 6: Continual learning results across all users. LoRA fine-tuning on accumulated data progressively improves classification accuracy from approximately 20% to over 50%, demonstrating the effectiveness of continual adaptation.

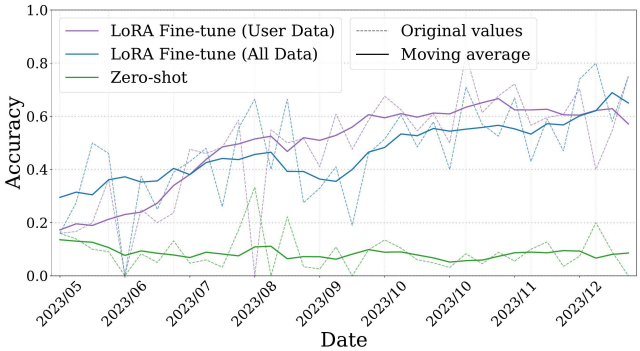

Figure 7: User-specific LoRA adaptation results for User A (a heavy logger). Personalized fine-tuning yields approximately 50 percentage points of accuracy improvement over the base model, demonstrating the value of user-aware adaptation.

Table 5: F1 score, Precision@1 (P@1), and Recall@K (R@1, R@3, R@5) for Qwen2.5-VL-7B-Instruct on the description-guided classification task.

	F1	P@1	R@1	R@3	R@5
w/o description	0.1594	0.1019	**0.3656**	**0.4590**	**0.4781**
w/ description	**0.2304**	**0.1788**	0.3240	0.4032	0.4336

management: individual users tend to eat a relatively consistent set of dishes, and a model tailored to their specific dietary patterns can achieve much higher accuracy than a one-size-fits-all approach.

5.4 Description-Guided Classification (Task 3)

Table 5 compares Qwen2.5-VL-7B performance with and without dish descriptions. Adding textual descriptions improves F1 from 0.159 to 0.230 (+44.7%) and P@1 from 0.102 to 0.179 (+75.5%), indicating that contextual information helps the model make more precise predictions. However, Recall@1 decreases from 0.366 to 0.324, and similar drops are observed for Recall@3 and Recall@5. This trade-off suggests that descriptions help the model focus on the correct answer when it would otherwise be uncertain, but can also mislead it when the description does not match the visual evidence.

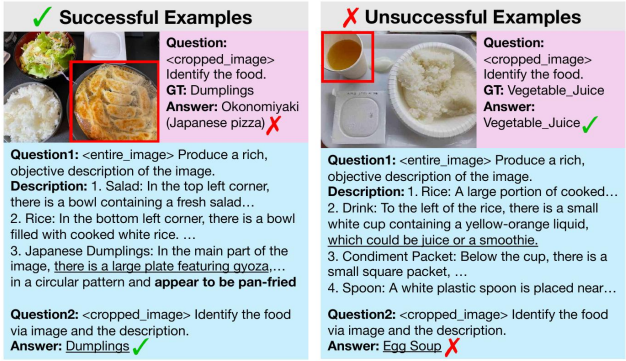

Figure 8: Examples of description-guided classification. Descriptions can help disambiguate visually similar dishes (left), but may also introduce errors when visual and textual signals conflict (right).

A detailed error analysis reveals that descriptions correct 687 previously wrong predictions but introduce 1,222 new errors. Figure 8 illustrates this dual effect: descriptions are most helpful for distinguishing between visually similar dishes that have distinct ingredient lists, but can cause errors when the visual appearance strongly conflicts with the textual description or when descriptions are overly generic.

These results suggest that effectively leveraging contextual information remains an open challenge, requiring better integration strategies that can weigh visual and textual evidence appropriately.

6 Conclusion

We presented FoodLogAthl-218, a food image benchmark dataset constructed from real dietary management application logs. Through a rigorous six-step pipeline combining automated CLIP-based filtering, manual review, and GPT-4o-assisted label consolidation, we produced a dataset of 6,925 images with 14,349 dish bounding boxes across 218 classes. Distribution analysis confirms that our dataset exhibits greater intra-class diversity and inter-class overlap than traditional web-sourced benchmarks, reflecting the genuine challenges of real-world dietary image recognition.

We defined three benchmark tasks addressing practical needs in dietary management. Our experiments reveal that current vision–language models struggle with real dietary images (best Recall@1: 42.4%), but that continual user-specific adaptation (+50pp accuracy) and description-guided recognition (+44.7% F1) offer promising avenues for improvement. We hope FoodLogAthl-218 will serve as a realistic testbed for advancing food image recognition toward practical deployment in dietary management systems.

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